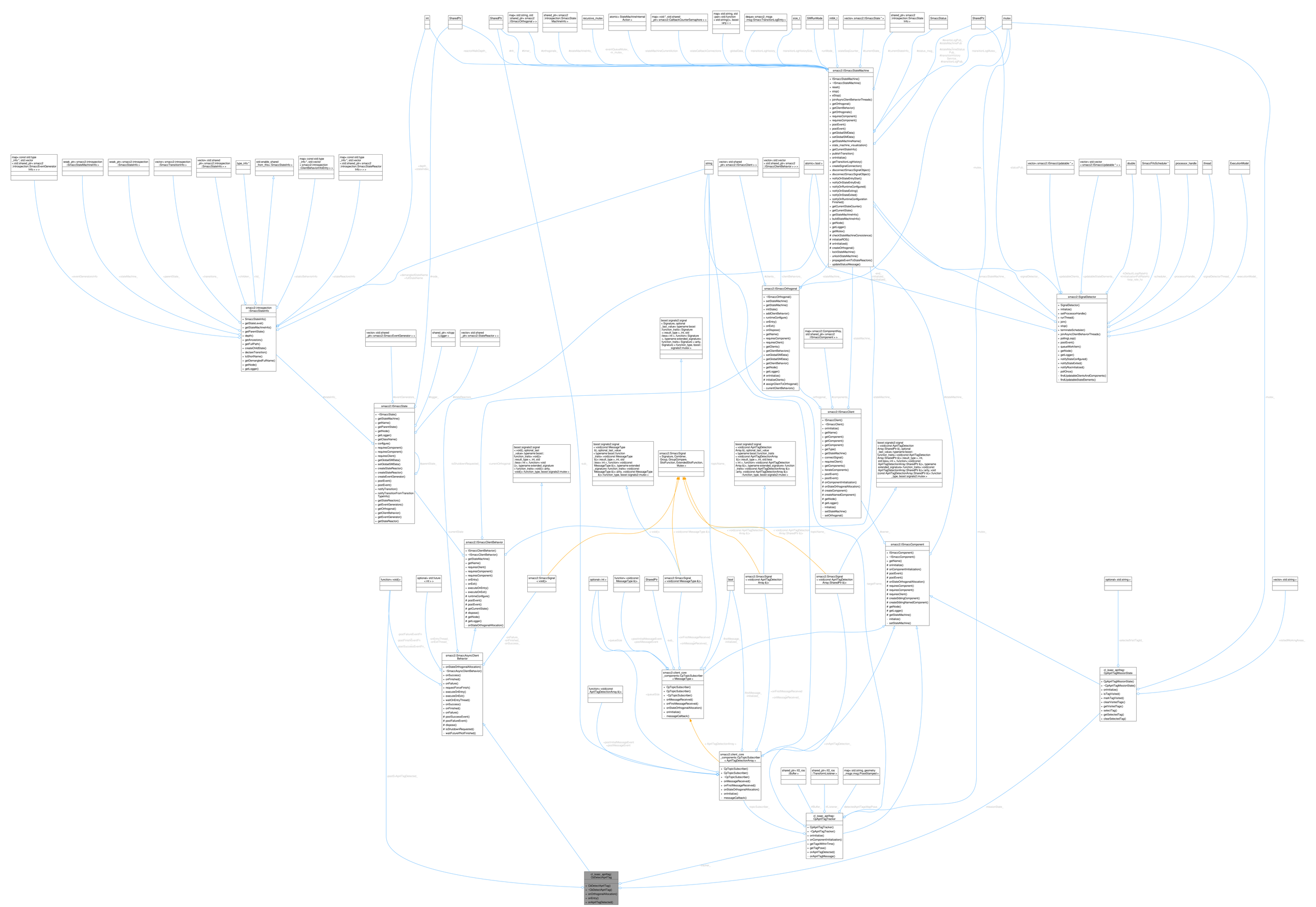
int
SharedPtr
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
deque< smacc2_msgs ::msg::SmaccTransitionLogEntry >
size_t
SMRunMode
int64_t
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
SharedPtr
mutex
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection ::SmaccStateInfo >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::ClientBehaviorWorkItEntry > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
string
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
atomic< bool >
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
double
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional _last_value= typename boost ::function_traits< Signature >::result_type, int= 0, class= std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature>::function_type, boost:: signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
vector< std::shared _ptr< smacc2::ISmaccEventGenerator > >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ oninitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < void(const MessageType &), optional_last_value < typename boost::function_traits < void(const MessageType &)>::result_type >, int, std::less< int >, function< void(const MessageType &)>, typename extended _signature< function_traits< void(const MessageType &)>::arity, void(const MessageType &)>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const AprilTagDetection Array &), optional_last_value < typename boost::function_traits < void(const AprilTagDetectionArray &)>::result_type >, int, std::less< int >, function< void(const AprilTagDetection Array &)>, typename extended _signature< function_traits< void(const AprilTagDetectionArray &)>::arity, void(const AprilTagDetectionArray &)>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const AprilTagDetection Array::ConstSharedPtr &), optional_last_value < typename boost::function_traits < void(const AprilTagDetectionArray::ConstSharedPtr &)>::result_type >, int, std::less< int >, function< void(const AprilTagDetectionArray::ConstSharedPtr &)>, typename extended_signature< function_traits< void(const AprilTagDetectionArray::ConstSharedPtr &)>::arity, void(const AprilTagDetectionArray::ConstSharedPtr &)>::function _type, boost::signals2::mutex >
boost::signals2::signal < void(), optional_last _value< typename boost ::function_traits< void()const>::result_type >, int, std::less< int >, function< void()const> , typename extended_signature< function_traits< void()const>::arity, void()const>::function_type, boost::signals2::mutex >
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# oninitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
smacc2::client_core _components::CpTopicSubscriber < MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ oninitialize()
- messageCallback()
smacc2::client_core _components::CpTopicSubscriber < AprilTagDetectionArray >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ oninitialize()
- messageCallback()
cl_isaac_apriltag::CpAprilTagTracker
+ CpAprilTagTracker()
+ ~CpAprilTagTracker()
+ oninitialize()
+ onComponentInitialization()
+ getTagsWithinTime()
+ getTagPose()
+ onAprilTagDetected()
+ onAprilTagMessage()
cl_isaac_apriltag::CpAprilTagMissionState
+ CpAprilTagMissionState()
+ ~CpAprilTagMissionState()
+ oninitialize()
+ isTagVisited()
+ markTagAsVisited()
+ clearVisitedTags()
+ getVisitedTags()
+ selectTag()
+ getSelectedTag()
+ clearSelectedTag()
cl_isaac_apriltag::CbDetectAprilTag
+ CbDetectAprilTag()
+ ~CbDetectAprilTag()
+ onOrthogonalAllocation()
+ onEntry()
+ onAprilTagDetected()
function< void()>
optional< std::future < int > >
smacc2::SmaccSignal < void()>
optional< int >
function< void(const MessageType &)>
SharedPtr
smacc2::SmaccSignal < void(const MessageType &)>
bool
smacc2::SmaccSignal < void(const AprilTagDetection Array &)>
smacc2::SmaccSignal < void(const AprilTagDetection Array::ConstSharedPtr &)>
optional< std::string >
vector< std::string >
function< void(const AprilTagDetectionArray &)>
shared_ptr< tf2_ros ::Buffer >
shared_ptr< tf2_ros ::TransformListener >
map< std::string, geometry _msgs::msg::PoseStamped >
+depth_
+stateIndex_
+eventGenerators_
+stateMachine_
+parentState_
+transitions_
+children_
+id_
+staticBehaviorInfo
+stateReactorsInfo
+demangledStateName
+fullStateName
+node_
+stateInfo_
+logger_
+stateReactors_
+reactorWalkDepth_
+sm_
+smac...
+orthogonals_
+stateMachineInfo_
+eventQueueMutex_ -m_mutex_
+stateMachineCurrentAction
+stateCallbackConnections
+globalData_
+transitionLogHistory_
+transitionLogHistorySize_
+runMode_
+stateSeqCounter_
+currentState_
+currentStateInfo_
+status_msg_
+eventsLogPub_ +stateMachinePub_
+stateMachineStatus Pub_ +transitionHistory Service_ +transitionLogPub_
+transitionLogMutex_
+statusPub_
+updatableClients_
+updatableStateElements_
+kDefaultLoopRateHz +kInitializationPollRateHz loop_rate_hz
+scheduler_
+processorHandle_
+signalDetectorThread_
+executionModel_
+mutex_
+clients_
+clientBehaviors_
+stateMachine_
+end_ +initialized_ +initialized_
+orthogonal_
+components_
+owner_
+targetName_
+topicName_
+sub_
+currentOrthogonal_
+isShutdownRequested_
+currentState_
+parentState_
+void()>
+< void(const MessageType &)>
+< void(const AprilTagDetection Array &)>
+< void(const AprilTagDetection Array::ConstSharedPtr &)>
+queueSize_
+postInitialMessageEvent +postMessageEvent
+onFirstMessageReceived_
+onMessageReceived_
+firstMessage_ +initialized_
+aprilTagDetectionArray_
+postInitialMessageEvent +postMessageEvent
+tfBuffer_
+tfListener_
+detectedAprilTagsMapPoses_
+topicSubscriber_
+aprilTagDetection_
+missionState_
+watcher_
+selectedVisitTagId_
+visitedWorkingAreas_
postFailureEventFn_
postFinishEventFn_
postSuccessEventFn_
onEntryThread_ onExitThread_
onFailure_ onFinished_ onSuccess_
postEvtAprilTagDetected_